

(847) 287-6910
Chicago, IL
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Mercedes Sandu

Software Engineer

Portfolio: mercedessandu.com
github.com/mercedes-sandu
linkedin.com/in/mercedes-sandu

Creative and motivated software, web, and game developer seeking to apply knowledge from both startup and industry experiences toward a dynamic and detail-oriented frontend, full-stack, or design position.

EDUCATION

Master of Science, Northwestern University
Computer Science, Thesis Track, GPA 4.0

March 2022 — June 2024

Bachelor of Arts, Northwestern University

September 2020 — June 2024

Mathematics and Computer Science Double Major, Chemistry Minor, GPA 3.87

TECHNICAL EXPERIENCE

ITPDP PROJECT MANAGER / M365 COPILOT AND COMMERCIAL DELIVERY

July 2025 — Present

Abbott

Chicago, Illinois

- Led planning and execution of the CDS M365 Copilot Evaluation Pilot for 103 licenses, achieving 90%+ adoption and identifying high-value use cases
- Developed a 28-slide charter presentation defining scope, success criteria, evaluation timeline, and communication plan for the pilot
- Designed and implemented a Power BI dashboard integrated with Microsoft Lists to track adoption, productivity KPIs, and gamification leaderboards
- Delivered weekly status reports and infographics to senior leadership, providing actionable insights on Copilot usage and time savings
- Organized and facilitated engagement events including training sessions, agent-athon/prompt-athon, mini hackathon, and Copilot Studio workshops with Microsoft
- Managed a team of six coaches, enabling effective cohort engagement and bi-weekly check-ins to drive participation and learning
- Conducted sentiment analysis surveys to measure user experience and inform evaluation outcomes.
- Presented evaluation kickoff and wrap-up at CDS Town Halls to audiences of 80+, showcasing results and strategic recommendations
- Achieved measurable impact: 200+ use cases submitted, 14,500+ hours saved annually, and 10,000 prompts written in November
- Delivered comprehensive final evaluation report and presentation, influencing future Copilot adoption across Abbott divisions
- Produced and refined six executive-level Items of Interest reports, ensuring CIO-ready status updates for CDS leadership
- Created and maintained six monthly portfolio review decks, analyzing project data and communicating risks to leadership
- Supported CDS Automation Maturity/Enhancement Factory initiative by designing surveys, performing data analysis, and creating executive summaries
- Coordinated CDS Sweep2 cybersecurity remediation plans with asset owners, ensuring compliance and risk mitigation

ITPDP DATA SCIENTIST / CRM DEMAND PLANNING

January 2025 — July 2025

Abbott

Chicago, Illinois

- Translated and optimized 15 R-based forecasting models into Python (Auto Arima, BATS, Croston, ETS, Holt, Holt-Winters, Moving Average, Prophet, Random Walk, STLF, SNaive, SPLine, TBATS, TSLM, and ThetaF)
- Developed modular, parameterized Python scripts integrated with GitHub for maintainability and scalability
- Built a robust forecasting pipeline in Azure ML Studio, reducing runtime from 40+ hours to under 4 hours
- Engineered scripts to compute and log key model metrics (MAE, MAPE, runtime, error tracking) for performance comparison
- Collaborated with cross-functional team members to parallelize and scale model execution across 12,000+ datasets
- Authored 8 unit tests to validate data integrity, forecast accuracy, and pipeline reliability
- Implemented comprehensive error handling and logging to ensure uninterrupted pipeline execution and traceability
- Designed and developed a new Power BI dashboard page for sales trends per quarter with custom DAX measures and interactive visualizations
- Created a 13-page user guide and a 3-minute demo video to support dashboard adoption by demand planners
- Presented project outcomes and technical learnings to senior leadership and the DS&E team
- Led planning and execution of Q1 and Q2 ITPDP Staff Meetings, including agenda creation, guest speaker coordination, and engagement initiatives
- Directed the design and launch of the 2025 BTS Global Data & AI Tech Summit SharePoint site, including Figma prototyping and post-event updates; reached over 8400 users
- Demonstrated strong leadership, collaboration, and technical versatility across backend, frontend, and organizational initiatives.

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ITPDP FRONTEND WEB ENGINEER / ABBOTT DESIGN SYSTEM

July 2024 — January 2025

Abbott

Chicago, Illinois

- Contributed to the Abbott Design System by developing 11 new web components using React, TypeScript, and MaterialUI, exceeding the initial goal of three components
- Created comprehensive documentation for each component using Storybook, detailing different variants and usage guidelines
- Updated 10 existing components to ensure compliance with original designs and requirements, collaborating closely with designers and lead developers
- Implemented support for theming and design tokens, enabling easier customization of components across different teams
- Developed a demo for testing components using Storybook's built-in features, enhancing the testing process for the Form component
- Installed and configured Playwright for end-to-end testing, writing demo tests for Abbott's CES website
- Reviewed 29 pull requests on GitHub, providing constructive feedback and ensuring code quality and functionality
- Facilitated onboarding and training for new React developers, outlining project structure, tech stack, and development pipeline
- Advocated for the use of Storybook across Abbott web products, creating visual documentation and installation guides
- Completed a 13-hour Playwright testing course, integrating Playwright into the project and writing end-to-end tests to improve testing capabilities

LEAD FRONTEND ENGINEER / INTERVALLIC, SONGLYBOT, OVERTURE.GAMES

June 2022 — November 2023

Overture Games

Chicago, Illinois

- Delivered and shipped Intervallic game on itch.io and Steam after just over a year of development
- Advised and guided team throughout the design and programming process of Intervallic using Unity and C#
- Delegated relevant tasks to team members with differentiated experience and roles on both development and art teams
- Fostered and led communication between the lead designer, the art team, and the development team
- Engineered and implemented primary game mechanics, including player movement, level progression, movement objects, accuracy detection, and Midi-to-game-object level generation
- Developed systems for Midi instrument, computer keyboard, and touchscreen input
- Built data structures to represent musical concepts such as key signatures, intervals, chromatic movement, and scales
- Led, designed, and programmed major UI/UX projects for visual aesthetic and frontend development of Intervallic game
- Developed SonglyBot, a Discord Python bot used to foster and engage with a community for the company through a Discord server allowing members to play fun minigames and have thoughtful conversations
- Developed and created content for Overture Games website using Svelte, TypeScript, HTML, and SCSS
- Designed and created in-game art for UI and other assets
- Conducted testing sessions for Intervallic game with hundreds of potential consumers and schools
- Mentored and pair programmed with new developers and artists joining the team during their onboarding
- Wrote technical and game design documentation for gameplay mechanics and systems developed using Confluence

GRADUATE RESEARCHER / AI & NARRATIVE RESEARCH, IMAGINARIUM, CATSAT

March 2022 — June 2024

Northwestern University Computer Science Department

Evanston, Illinois

- Studied and documented the uses of propositional and formal logic in generative software and game artificial intelligence
- Analyzed Dr. Ian Horswill's experimental generative language, Imaginarium, with the intent to expand its logical implications
- Experimented with Imaginarium project and extended Imaginarium codebase to discover its technical and logical shortcomings and planned improvements
- Designed, implemented, and tested a constraint-based SAT solver for constraints imposed on randomly generated graphs, expanding upon Dr. Horswill's project, CatSAT
- Created and documented constraints to impose on graphs in CatSAT, such as connected graphs and connected nodes
- Utilized and implemented data structures such as UnionFind, spanning trees, graphs, and paths
- Visualized and verified test cases by writing code using C# testing libraries and creating Graphviz .dot files
- Wrote and defended a Masters thesis presenting work done

FRONTEND DEVELOPMENT IT INTERN / KNOWme

June 2023 — September 2023

Abbott

Chicago, Illinois

- Designed and implemented components for ecommerce React webapp using TypeScript, HTML, and CSS
- Cleared backlog of ADA audits and documented changes and passing Lighthouse tests
- Updated webapp components to be accessible with AA-standard colors, tabbed navigation focus states, and accessibility labels
- Collaborated across multiple teams to develop different parts of the webapp: pre-purchase (product listings and pre-screening questions), post-purchase (test results), and component library
- Completed tasks in an agile sprint-based workflow
- Redesigned and implemented PowerBI dashboards and PowerApps using Figma for intern showcase project

UNDERGRADUATE RESEARCHER / DISCRETE GEOMETRY RESEARCH

January 2021 — June 2023

Northwestern University Mathematics Department

Evanston, Illinois

- Studied the previous work of Dr. Shuyi Weng and Dr. Laura DeMarco on the folding of two-dimensional polygons into three-dimensional shapes
- Conducted case studies on different shapes and used findings and implemented code to write formal proofs
- Wrote code in $\text{\LaTeX}$ and Mathematica to analyze case studies and create interactable figures
- Wrote a formal mathematics academic paper discussing main findings of research on polygons and three-dimensional shapes
- Published academic paper in *Involve Journal*, written with mentor
- Presented research findings to hundreds of educated enthusiasts at Northwestern University
- Presented research findings to undergraduate students, graduate students, and professors at national Joint Mathematics Meetings

SKILLS

Tools and Languages	C#, Unity, Java, Python, HTML, CSS, TypeScript, React, Figma, $\text{\LaTeX}$, Blender, Git, Discord API, Jira
Quantitative Research	Proof Writing, Discrete Geometry, Artificial Intelligence, Logic, Mathematica, MatLab
Communication	English, Romanian, Spanish

AWARDS, PRESENTATIONS, & PUBLICATIONS

Abbott DVP Award, CRM Demand Planner Tool	June 2025
<i>Closed cap condition under the cap construction algorithm</i> , Involve Journal	December 2024
<i>Intervallig</i> , Steam and itch.io	October 2023
Joint Mathematics Meetings Undergraduate Oral Presentation	January 2023
Financial Award for The Garage Jumpstart Competition Finalist	August 2022
Summer Undergraduate Research Grant	Summer 2022
Undergraduate Research Exposition Presentation	May 2021
Northwestern Undergraduate Mathematical Society Presentation	May 2021
Dean's List	Spring 2024, Winter 2023, Fall 2022, Spring 2022, Winter 2022, Fall 2021, Winter 2021, Fall 2020